



Basic Details of the Team and Problem Statement

Ministry/Organization Name/Student Innovation: **Indian Space Research Organisation (ISRO)**

PS Code: **SIH1521**

Problem Statement Title: **Development of Explainable AI (XAI) based model for prediction of heavy /high impact rain events using satellite data**

Team Name: **ByteMasters**

Team Leader Name: **Pranay Saha**

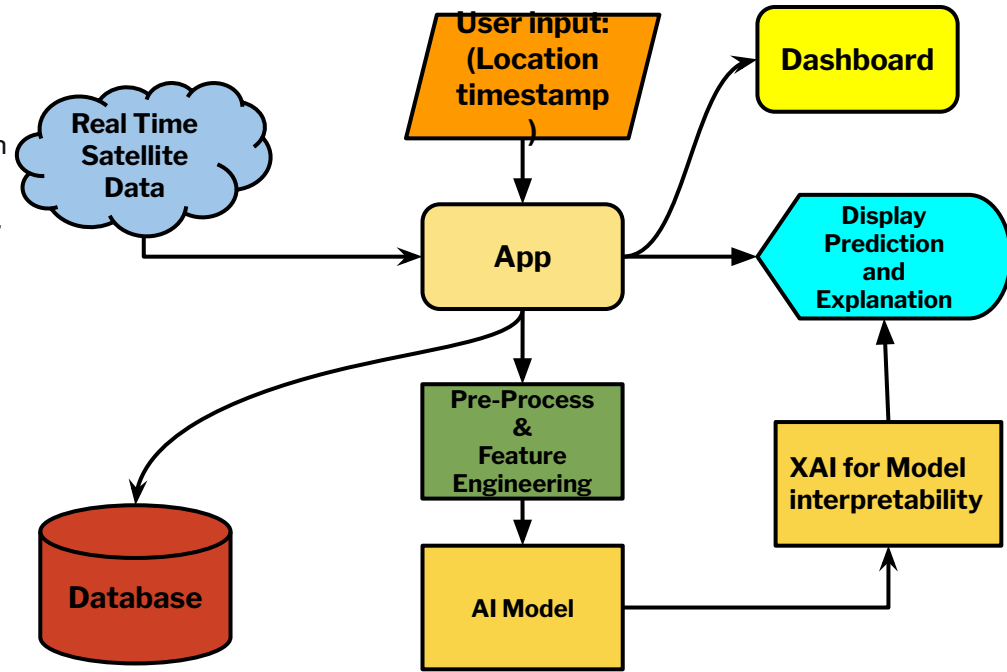
Institute Code (AISHE): **C-6192**

Institute Name: **Institute of Engineering and Management**

Theme Name: **Space Technology**

Idea/Approach Details

- Analyze and process data from geostationary satellites such as INSAT-3D/3DR. **Capture multispectral imagery** consists of cloud cover imagery, temperature profiles from sounders, humidity levels, wind patterns, and historical rainfall data.
- Implement AI models like CNNs or LSTM for **predictive modeling** Fuse real-time satellite data, allowing the model to incorporate current conditions into forecasts.
- Enhance **interpretability** using techniques like **SHAP** or **LIME**, monitor predictions to track and Identify potential model failures, considering factors like data quality, sudden weather changes, and calibration issues.
- Create a **web application** that allows users to input parameters (e.g., date, time, location) and receive predictions for heavy precipitation events. Display model predictions and explanations in a user-friendly manner.



TECHNOLOGY STACK:



Idea/Approach Details

USE CASES:

- **Severe Weather Forecasting:** The system enables the prediction of severe weather events such as thunderstorms, hurricanes, and heavy rainfall, allowing for timely alerts and disaster preparedness.
- **Flood Prediction and Management:** Analyzing rainfall data and monitoring river levels, it aids in flood prediction and management, helping authorities and communities take preventive measures.
- **Agricultural Planning:** Farmers can use rainfall predictions to make informed decisions regarding crop planting, irrigation, and harvest schedules, enhancing agricultural productivity.
- **Climate Research:** The system's data contributes to climate research by providing insights into long-term rainfall patterns and trends, facilitating climate change studies.
- **Disaster Response:** Rapid access to real-time weather data assists disaster response teams in planning and executing relief efforts during natural disasters.

DEPENDENCIES / SHOWSTOPPERS:

- **Data Availability:** The availability of timely and accurate satellite data, like INSAT-3D/3DR, is a primary dependency. A lack of access to this data can hinder the entire system.
- **Data Quality:** Ensuring the quality and reliability of satellite data is essential. Any issues with data accuracy or calibration can impact the system's performance.
- **Data Privacy and Security Concerns:** Compliance with data privacy regulations and ensuring the security of sensitive weather data is critical. Violations can lead to legal and ethical showstoppers.
- **Limited Access to Real-Time Data:** A lack of access to up-to-the-minute satellite data, particularly during adverse weather conditions, can hinder real-time predictions.
- **Domain Expertise:** Dependence on meteorologists and domain experts is critical for the interpretation of data and model performance evaluation.

Team Member Details

Team Leader Name: Pranay Saha

Branch (Btech/Mtech/PhD etc): BTech Stream (ECE, CSE etc): CSE(AIML) Year (I,II,III,IV): IV

Team Member 1 Name: Kulsum Kamal

Branch (Btech/Mtech/PhD etc): BTech Stream (ECE, CSE etc): CSBS Year (I,II,III,IV): IV

Team Member 2 Name: Prisati Bhattacharjee

Branch (Btech/Mtech/PhD etc): BTech Stream (ECE, CSE etc): CSE(AIML) Year (I,II,III,IV): IV

Team Member 3 Name: Sugoto Basu

Branch (Btech/Mtech/PhD etc): BTech Stream (ECE, CSE etc): CSE(AIML) Year (I,II,III,IV): IV

Team Member 4 Name: Jeet Chovatia

Branch (Btech/Mtech/PhD etc): BTech Stream (ECE, CSE etc): CSE(AIML) Year (I,II,III,IV): IV

Team Member 5 Name: Subha Chakraborty

Branch (Btech/Mtech/PhD etc): BTech Stream (ECE, CSE etc): CSE(AIML) Year (I,II,III,IV): IV

Team Mentor 1 Name: Subhadip Chandra

Category: (in years): 14	Hackathon	Expertise: Applied Soft Computing, Machine Vision	Domain Experience
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